

Chapter 8 — Exercises

Multiple-Choice Questions

Wrapping up: the bigger picture

How to Use These Slides

- **12 multiple-choice** questions on Chapter 8 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: approaching a problem, humans in the loop, production, custom estimators, scaling, and where to go next.

Q1. Before running an algorithm, you should first:

- A. frame the problem and decide how to measure success
- B. pick the most complex model available
- C. collect as many features as possible
- D. deploy to production

Q2. The "what if I built the perfect model?" question helps assess:

- A. which algorithm is fastest
- B. whether the problem is worth solving (its business impact)
- C. the number of features to use
- D. the learning rate

Q3. "Humans in the loop" typically means:

- A. humans label every single prediction
- B. removing all automation
- C. routing uncertain or complex cases to a person
- D. training only on human-written data

Q4. A/B testing is a form of:

- A. offline evaluation on historical data
- B. cross-validation
- C. feature selection
- D. online (live) evaluation with real users

Q5. To build a scikit-learn-compatible transformer, inherit from:

- A. `BaseEstimator` and `TransformerMixin`
- B. `Pipeline` and `GridSearchCV`
- C. `numpy.ndarray`
- D. `StandardScaler`

Q6. Out-of-core learning is used when:

- A. the data has no labels
- B. the data doesn't fit in RAM, so it's streamed in chunks
- C. you need a GPU
- D. the model is a decision tree

Q7. For distributed learning across a cluster, a common tool is:

- A. matplotlib
- B. pandas
- C. spark with MLlib
- D. Jupyter

Q8. Probabilistic programming languages include:

- A. NumPy and SciPy
- B. R and MATLAB
- C. Go and Scala
- D. PyMC and Stan

Q9. For statistical modeling and inference in Python, consider:

- A. statsmodels
- B. matplotlib
- C. Pillow
- D. requests

Q10. Ranking is the kind of learning behind:

- A. image classification
- B. search engines
- C. clustering
- D. data scaling

Q11. Kaggle and OpenML are mainly useful for:

- A. deploying production systems
- B. writing documentation
- C. practicing on datasets and competitions
- D. scaling to clusters

Q12. Moving a prototype to production often involves:

- A. never changing the code
- B. using only Jupyter notebooks
- C. ignoring runtime and memory
- D. reimplementing it in a high-performance language (Go/Scala/C++/Java)

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	A	Start by framing the problem and defining success
2	B	It gauges the business impact — is it worth building?
3	C	Automate simple cases, route hard ones to humans
4	D	A/B testing is online / live evaluation
5	A	Inherit <code>BaseEstimator</code> + <code>TransformerMixin</code>
6	B	Out-of-core streams data in chunks (larger than RAM)

Answers 7–12

Q	Ans	Why
7	C	spark + MLlib distributes learning over a cluster
8	D	PyMC and Stan are probabilistic programming tools
9	A	statsmodels targets statistical modeling & inference
10	B	Ranking powers search engines
11	C	Kaggle / OpenML → practice datasets & competitions
12	D	Production often means reimplementing in a fast language

Well done!

You've completed the course exercises!